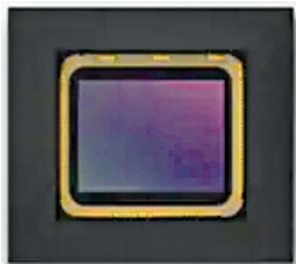


Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

Image sensor for automotive applications

The ISOCELL Auto 4AC, an automotive image sensor, offers advanced 120-decibel high dynamic range and LED flicker mitigation for surround-view monitors



or rear-view cameras in high-definition resolution (1280 x 960). It gives a safer driving experience with an enhanced field of view for the driver thanks to CornerPixel technology, which features a specialised pixel structure that mitigates LED light over 90Hz. Within a single pixel area, it embeds two photodiodes, one 3.0µm-pixel for viewing low light images and a 1.0µm-pixel placed at the corner of the big pixel for brighter environments. The two photodiodes simultaneously capture images in different scenarios, thus allowing smoother transitions between dark and bright areas while preserving more details of the road ahead.

Samsung
www.samsung.com

Self-charging temperature and humidity sensor

The HADAR sensor is a wireless temperature and humidity sensor that uses LoRaWAN technology. Powered by a light harvesting cell, it works absolutely maintenance-free; no batteries need to be replaced throughout its lifecycle. An integrated energy store allows it to

operate for several days in total darkness. The acquired information can be used for optimising heating/cooling and also to supervise the humidity to avoid



any damage. With respect to security, the hardware based security architecture provides secure boot, encrypted firmware, and safe key storage, making sure correct data is fed without a risk of information leakage or fraud. Operating temperature is from -20°C to +65°C.

Pollux
www.pllx.se

Battery chargers for wearable, medical and smart sensor products

The AEM10900 and AEM10300 (solar) and AEM30300 (vibration and RF) PMICs are fully dedicated to the charging function, addressing the need for streamlined power management solutions for energy harvesting hardware. This allows them to be simply employed



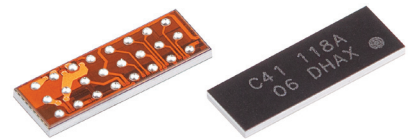
in situations where there are space or cost constraints. The devices achieve zero quiescent current drawn from the battery—meaning that if energy harvesting stops for a prolonged period of time, the energy stored into the battery will not be wasted supplying the PMIC. AEM10900 has a 250mV cold start capability, which means it can commence with charging the battery even when light intensities are extremely low.

AEM10300 and AEM30300 PMICs both have built-in ultra-low-power DC-DC converters supporting operation over an input voltage range that covers 100mV to 4.5V.

e-peas
www.e-peas.com

Driver ICs for shape memory alloy applications

The ultra-small and high-performance CM401 consists of built-in micro-controller, flash memory, SRAM and proprietary power delivery, position



sensing and precision control logic, specially tailored for driving SMA based lens shift OIS actuators and other shape memory alloy based applications such as 3D sensing, haptics, and more. The development also helps in maintaining short time to market—from conception all the way to mass production.

Cambridge Mechatronics
www.cambridgemechatronics.com

GPS technology with sensor fusion simplifies precise positioning

The ZED-F9R-02B has been designed for use in autonomous and industrial applications that require simple and efficient implementation, and where rapid access to highly accurate positioning data is key, even in challenging signal environments such as dense cities. Applying real-time kinematic techniques, the module can derive positioning data with centimetre-level accuracy. The high-precision GNSS measurements are fused with data from the module's